Alexander Beattie

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Education

Tufts University Medford, MA
Bachelor of Science, Mechanical Engineering GPA:3.33 May 2027
Relevant Coursework: Engineering Design I & II, Electronics & Controls I & II, Materials & Manufacturing I & II, Mechanics I & II, Thermal Fluid Systems I & II, Instrumentation & Experiments

Engineering Experience

NOLOP FAST Facility Medford, MA
Fabrication Staff Jan 2026 - Present

- Enable high-quality fabrication by advising students and researchers on material selection, tolerance stack-ups, and DFM best practices
- Support complex builds using 3D printing, laser cutting, and CNC milling equipment
- Enforce safety protocols and deliver hands-on training in a high-throughput makerspace setting

Derby Entrepreneurship Center, Tufts University Medford, MA
Teaching Assistant, ENT164: Intro to Making Aug 2024 - Present

- Guide 50+ students per semester through full design-build-test cycles of electromechanical systems
- Troubleshoot integration issues involving sensors, actuators, and microcontrollers under real project constraints
- Iterated course materials and microcontroller transitions across 4+ semesters to improve clarity and build success

Tufts Center for Engineering Education and Outreach (CEEO) Medford, MA
Teaching Fellow, Engineering Design Lab Summer 2025

- Led high school students in building robotic systems using Python, Raspberry Pi, and digital fabrication
- Designed and taught IoT and control system modules using Jupyter Notebooks
- Built demos, refined curriculum, and mentored project teams to develop fully integrated mechatronic systems

Leadership & Activities

Zeta Beta Tau Fraternity, Chapter President Nov 2025 - Present
F.I.T. Pre-Orientation, Peer Leader Aug 2024 - Present
Tufts x MIT Leadership Training Institute, Head Leadership Mentor Jan 2024 - Present

Projects

Detangulator 5000 Assistive Device
Designed a low-force positioning/actuation aid to help a client with cerebral palsy independently apply hair detangler.

Cooperative Payload Balancing Robot
Built a Raspberry Pi robot with load-cell sensing, op-amp/ADC signal conditioning, and a custom web UI; calibrated for various payloads (up to 1 kg) and repeatable torque-balanced placement on a seesaw while coordinating with a partner team.

Skills

Fabrication & Manufacturing: 3D Printing, Laser Cutting, CNC Machining, Rapid Prototyping
Hardware & Embedded Systems: Raspberry Pi, ESP32, GPIO interfacing, common sensors & actuators
Programming: Python, Jupyter Notebooks, MATLAB, LabVIEW
Software: CAD (Solidworks, Onshape, AutoCAD), Solidworks FEA, KiCAD, COMSOL Multiphysics
Interests: Tufts Climbing Team member, Photography, teaching/mentorship